

Individual Project

(Large-Scale Data Storage)

Car Price Analysis using Hadoop Ecosystem

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1. Introduction:

With the rapid increase in data generation across industries, organizations require efficient technologies to store, process, and analyze large datasets. Hadoop is one of the most widely used frameworks for distributed storage and parallel data processing.

This project focuses on analyzing a Car Price Dataset using multiple components of the Hadoop ecosystem, including HDFS, MapReduce, Hive, Pig, and HBase.

The project demonstrates how large datasets can be stored and processed in a distributed environment and how the processed data can be visualized using Python.

The primary objectives of the project are:

- To store large datasets using HDFS
- To process data using MapReduce
- To perform SQL-like analysis using Hive
- To perform data transformations using Pig
- To store structured data in HBase
- To create visualizations and dashboards using Python

2. Dataset Description:

The dataset used in this project contains information about various vehicles and their pricing details.

The dataset includes attributes such as:

- Car ID
- Brand
- Model
- Year
- Engine Size
- Fuel Type
- Transmission
- Mileage
- Condition
- Price

The dataset size is approximately 152 MB, making it suitable for big data processing using Hadoop.

3. Hadoop Environment Setup (Week 1):

The Hadoop environment was configured using the Hortonworks Data Platform (HDP) Sandbox.

The sandbox was installed using Oracle VirtualBox, which allowed the Hadoop cluster to run locally.

Steps performed:

- Installed Oracle VirtualBox
- Imported HDP Sandbox virtual machine
- Configured network settings
- Accessed the Hadoop environment through SSH
- Verified Hadoop services using Ambari

Important Hadoop services used in the project include:

- HDFS
- YARN
- MapReduce
- Hive
- Pig
- HBase

4. HDFS Data Storage (Week 2):

In this stage, the dataset was uploaded to the Hadoop Distributed File System (HDFS).

HDFS allows large datasets to be stored across multiple nodes, enabling distributed storage and high fault tolerance.

Steps performed:

- Created project directory in HDFS
- Uploaded the dataset
- Verified file storage
- Viewed dataset contents

Commands used:

```
hdfs dfs -mkdir /user/maria_dev/car_project
```

```
hdfs dfs -put car_price_dataset.csv /user/maria_dev/car_project
```

```
hdfs dfs -ls /user/maria_dev/car_project
```

```
hdfs dfs -cat /user/maria_dev/car_project/car_price_dataset.csv
```

5. MapReduce Processing (Week 3):

MapReduce was used to process the dataset and compute the average price of vehicles by brand.

The MapReduce program consists of two phases:

Map Phase

The mapper reads each record and outputs:

Brand → Price

Reduce Phase

The reducer groups values by brand and calculates the average price.

Example output:

BMW 54157

Audi 51953

Ford 51593

Honda 52050

Tesla 53475

Toyota 52078

Mercedes 53191

```
Audi      51953.4248098
BMW       54157.1143855
Ford      51593.2548127
Honda     52050.2839488
Mercedes          53191.090085
Tesla     53475.5474713
Toyota    52078.7282353
```

6. Hive Query Analysis (Week 4):

Apache Hive was used to perform structured queries on the dataset stored in HDFS.

Hive allows SQL-like queries to analyze large datasets efficiently.

Example queries executed:

Average price by brand

```
SELECT brand, AVG(price)
```

```
FROM car_prices
```

```
GROUP BY brand;
```

Top expensive vehicles

```
SELECT brand, model, price
```

```
FROM car_prices
```

```
ORDER BY price DESC
```

```
LIMIT 10;
```

7. Pig Data Processing (Week 5):

Apache Pig was used for data transformation and aggregation.

Pig provides a high-level scripting language that simplifies complex data processing tasks.

Example Pig script:

```
cars = LOAD '/user/maria_dev/car_project/car_price_dataset.csv'
      USING PigStorage(',')
      AS (id:int, brand:chararray, year:int, engine:float,
          fuel:chararray, transmission:chararray,
          mileage:int, condition:chararray, price:float, model:chararray);
grouped = GROUP cars BY brand;
avg_price = FOREACH grouped GENERATE group, AVG(cars.price);
DUMP avg_price;
```

8. HBase NoSQL Storage (Week 6):

In this stage, Apache HBase was used to store structured car data.

Table created:

Car_price_hbase

Example commands:

Create table

```
create 'car_price_hbase', 'basic', 'specs'
```

Insert data

```
put 'car_price_hbase', 'BMW_2018_1', 'basic:brand', 'BMW'
```

```
put 'car_price_hbase', 'BMW_2018_1', 'basic:price', '54157'
```

```
put 'car_price_hbase', 'BMW_2018_1', 'specs:engine', '3.0'
```

Scan table

```
scan 'car_price_hbase'
```

9. Data Visualization using Python (Week 7):

Python libraries Pandas and Matplotlib were used to create visualizations.

The following charts were generated:

- Average Price by Brand
- Fuel Type Distribution
- Transmission Distribution
- Car Condition Distribution
- Average Price by Year
- Top 5 Brands by Market Presence
- Price Range Distribution

Figure 1: Average Price by Brand

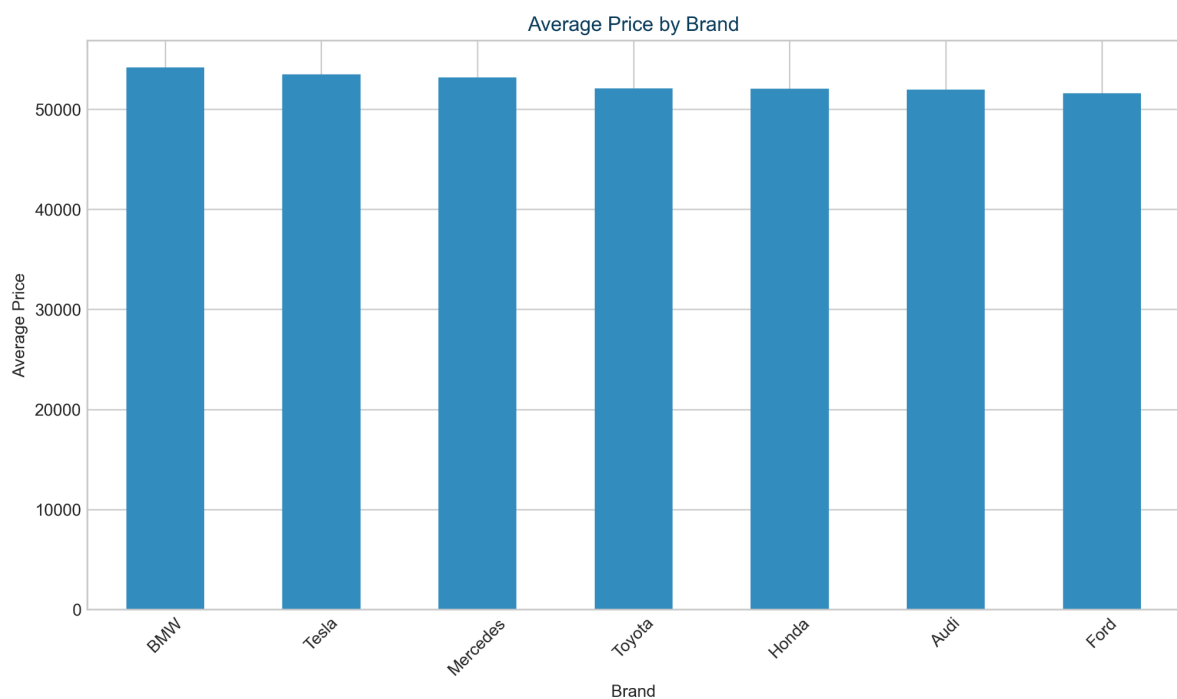


Figure 2: Fuel Type Distribution

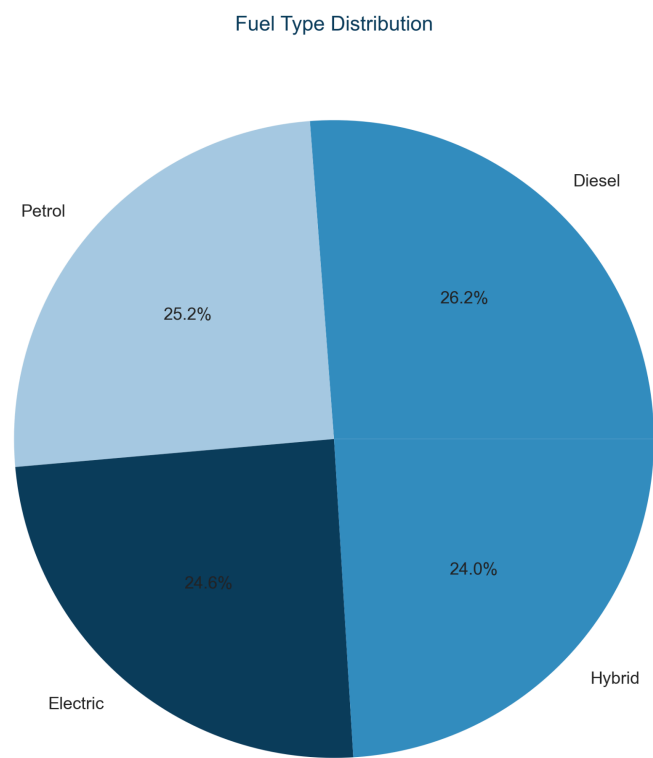


Figure 3: Transmission Distribution

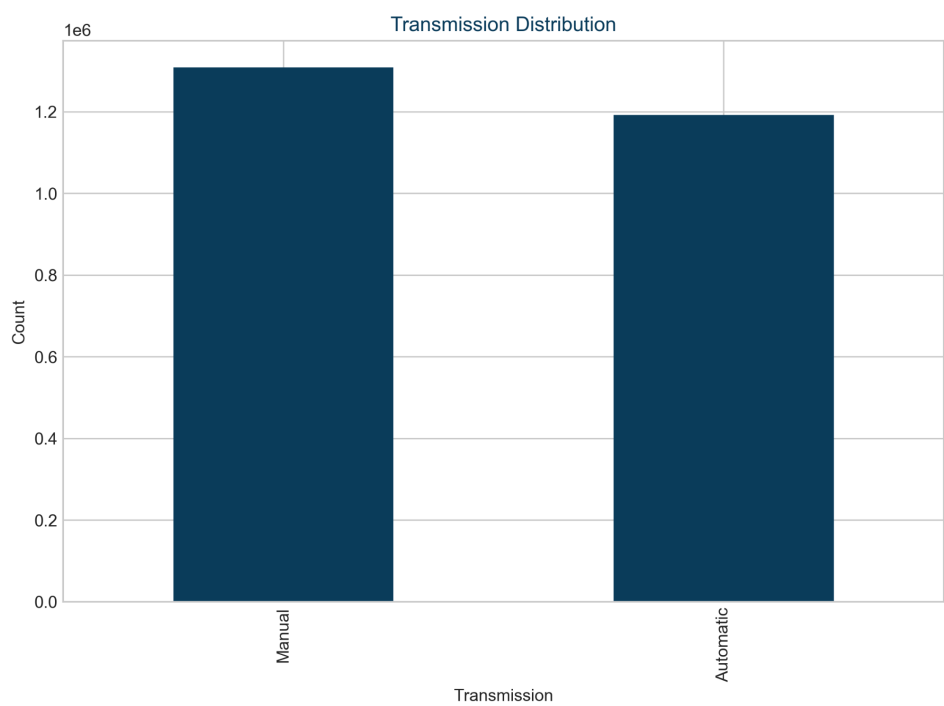


Figure 4: Car Condition Distribution

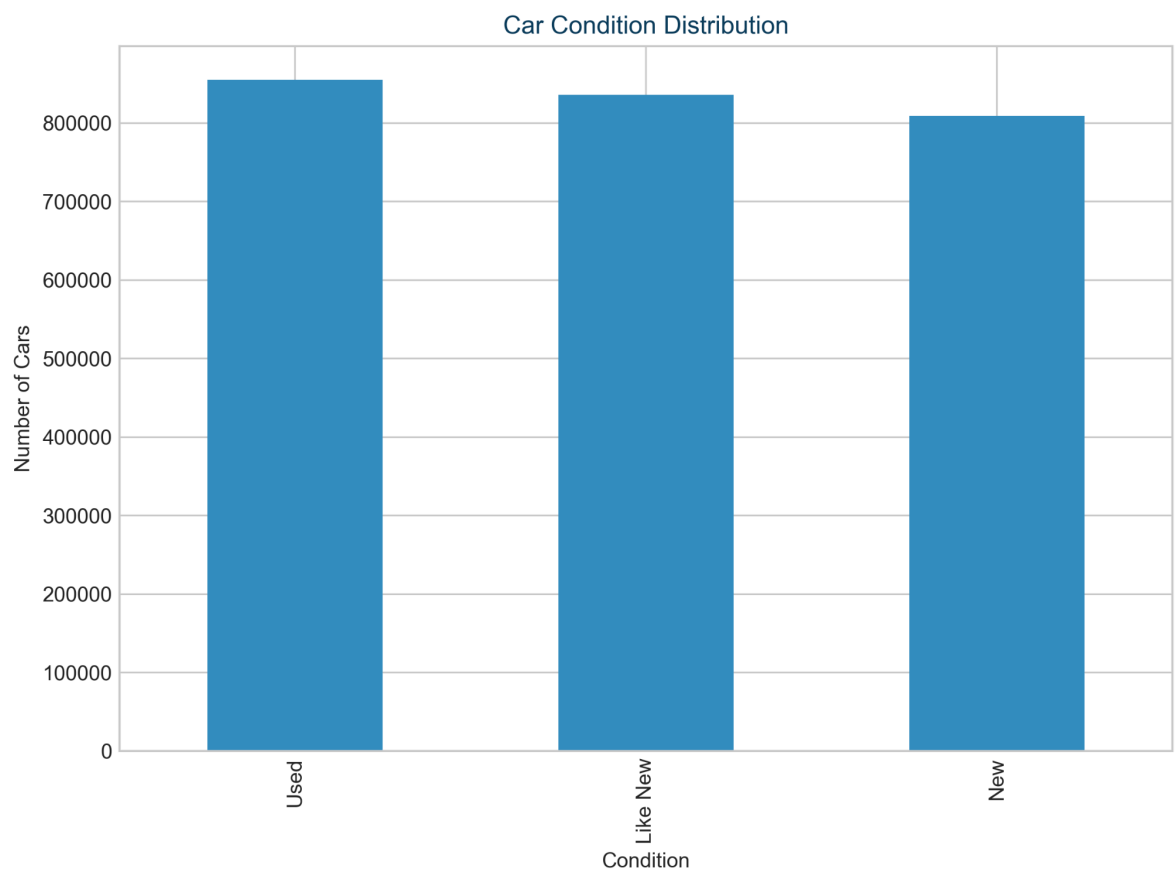


Figure 5: Average Price by Year

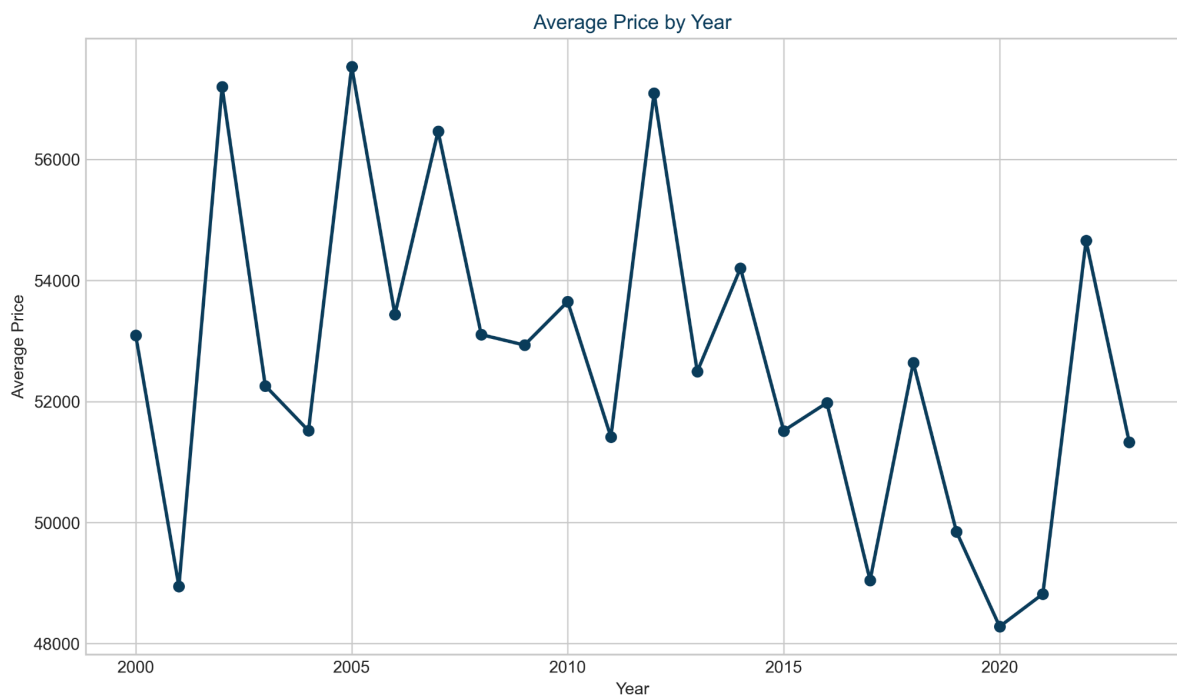


Figure 6: Top 5 Brands by Market Presence

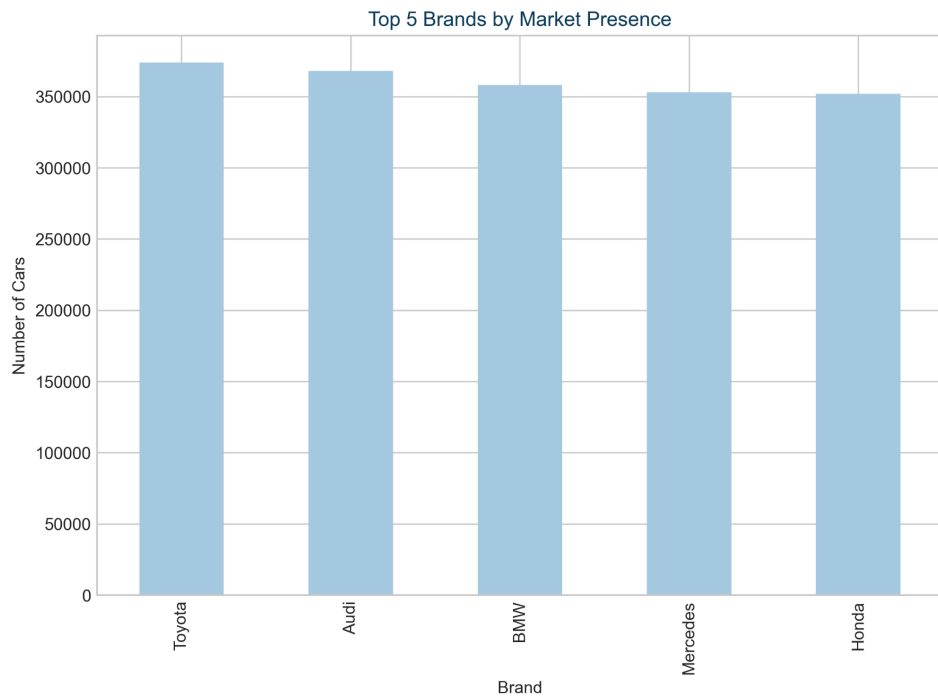
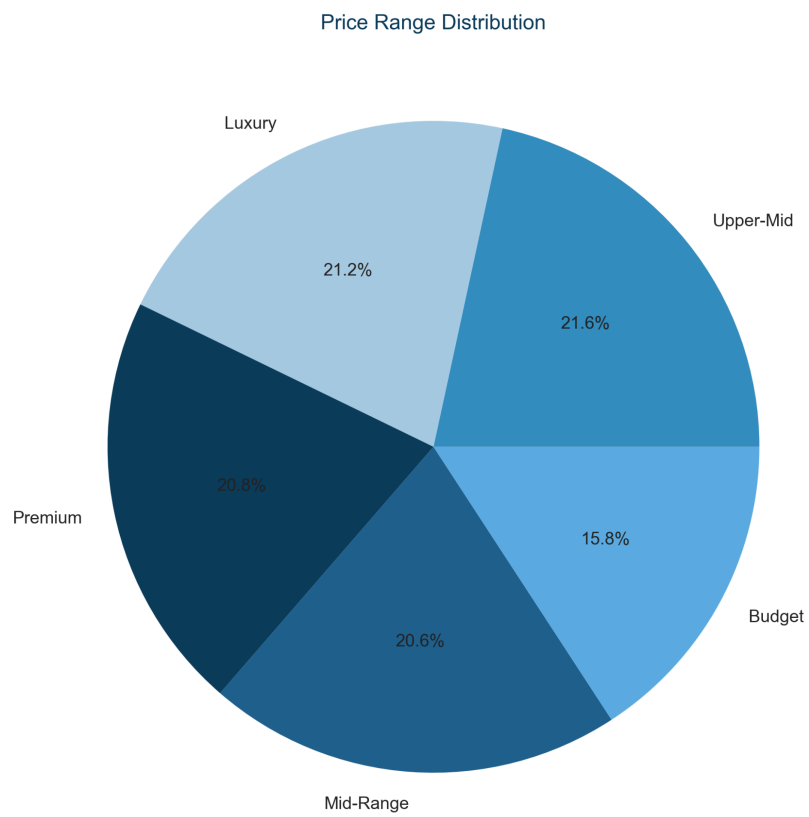


Figure 7: Price Range Distribution



10. Real-Time Data Streaming using Kafka (Week 7):

In this stage, Apache Kafka was used to simulate real-time data streaming for car price analysis. Unlike previous steps where static datasets were processed, Kafka enabled continuous data ingestion, making the system more realistic and closer to real-world applications.

Kafka is a distributed streaming platform that allows data to be produced and consumed in real time using topics.

Steps Performed:

- Kafka service was started using Ambari in the HDP Sandbox environment.
- A Kafka topic named car-price-topic was created to store streaming data.
- A producer was implemented using the Kafka console producer to send car data such as brand, year, and price into the topic.
- Example data streamed:

Toyota,2018,450000

Honda,2020,600000

BMW,2022,2500000

- A consumer was used to read the data from the Kafka topic in real time.
- The streamed data was then saved into a file (car_data.txt).
- The file was uploaded into HDFS for further processing and analysis using Hive.

Kafka Data Flow:

Producer → Kafka Topic → Consumer → File → HDFS → Hive

```
^C[maria_dev@sandbox-hdp bin]$ ./kafka-console-producer.sh \  
> --topic car-price-topic \  
> --broker-list sandbox-hdp.hortonworks.com:6667  
>Toyota,2018,450000  
>Honda,2020,600000  
>BMW,2022,2500000  
>
```

```
[maria_dev@sandbox-hdp bin]$ ./kafka-topics.sh --create \  
> --topic car-price-topic \  
> --zookeeper localhost:2181 \  
> --partitions 1 \  
> --replication-factor 1  
Created topic "car-price-topic".  
[maria_dev@sandbox-hdp bin]$ ./kafka-topics.sh --list --zookeeper localhost:2181  
ATLAS_ENTITIES  
__consumer_offsets  
car-price-topic  
[maria_dev@sandbox-hdp bin]$
```

```

maria_dev@sandbox-hdp:/usr/hdp/current/kafka-broker/bin
Using username "maria_dev".
maria_dev@127.0.0.1's password:
Last login: Mon Mar 30 12:32:21 2026 from 172.18.0.3
[maria_dev@sandbox-hdp ~]$ cd /usr/hdp/current/kafka-broker/bin
[maria_dev@sandbox-hdp bin]$ ./kafka-console-consumer.sh \
> --topic car-price-topic \
> --from-beginning \
> --zookeeper localhost:2181
Using the ConsoleConsumer with old consumer is deprecated and will be removed in a future major release. Consider using the new consumer by passing [bootstrap-server] instead of [zookeeper].
Toyota,2018,450000
Honda,2020,600000
BMW,2022,2500000

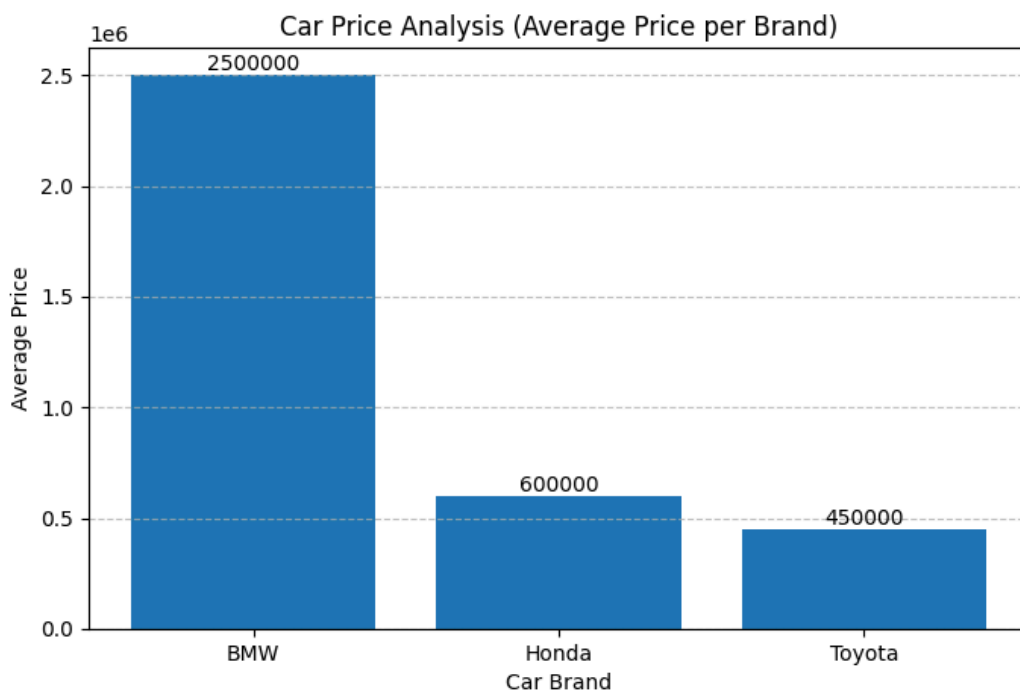
```

Outcome:

Using Kafka, real-time streaming of car data was successfully simulated. The data was consumed and stored into HDFS, and further analyzed using Hive queries. This demonstrated the integration of real-time data ingestion with batch processing in the Hadoop ecosystem.

Advantages of Using Kafka:

- Enables real-time data streaming
- High throughput and scalability
- Fault-tolerant and reliable
- Integrates easily with the Hadoop ecosystem.



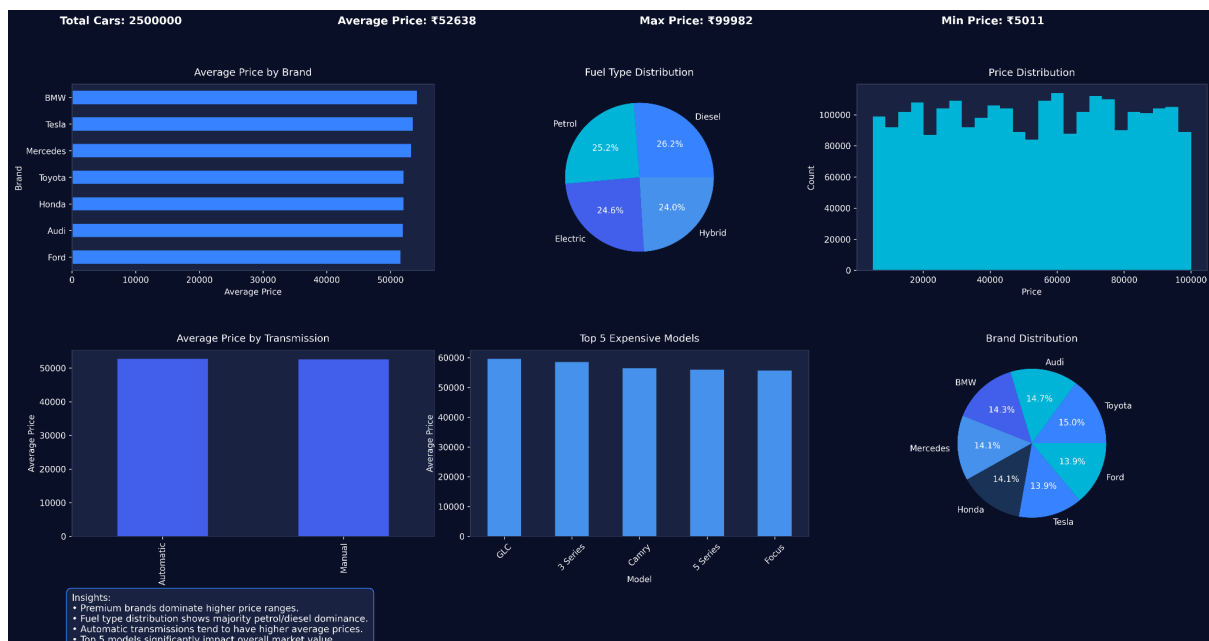
11. Dashboard Visualization:

A dashboard was created to display multiple charts together for better analysis.

The dashboard includes:

- Brand price comparison
- Fuel distribution
- Price histogram
- Transmission analysis
- Top expensive models
- Brand distribution

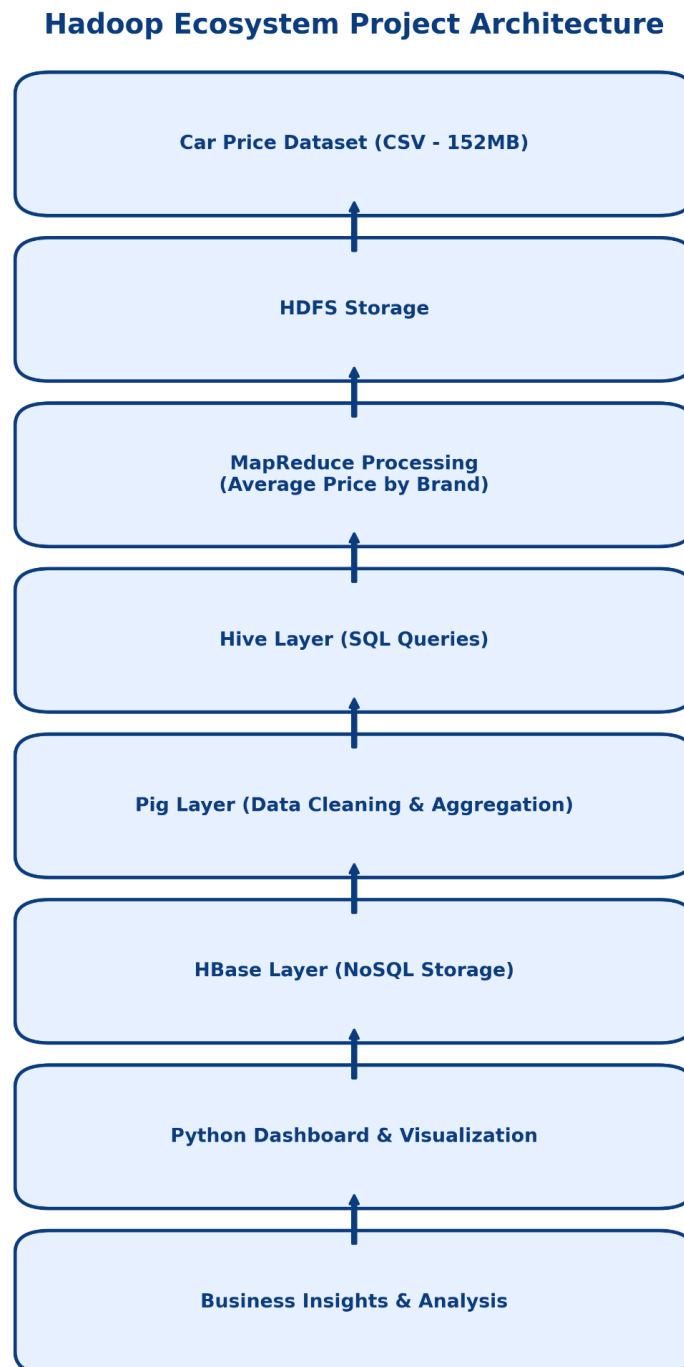
Figure 8: Car Price Analysis Dashboard



12. Hadoop Ecosystem Architecture:

The project architecture shows how the Hadoop ecosystem components interact within the project.

Workflow:



13. Key Insights:

From the analysis, the following insights were obtained:

- BMW, Tesla, and Mercedes show higher average car prices.
- Diesel vehicles have the largest share among fuel types.
- Manual transmission vehicles appear slightly more frequently than automatic vehicles.
- Used vehicles dominate the dataset.
- Toyota has the highest brand representation.
- Most vehicles fall within mid-range price categories.

14. Conclusion:

This project demonstrated how the Hadoop ecosystem can be used to store, process, and analyze large datasets.

Different tools such as HDFS, MapReduce, Hive, Pig, and HBase were successfully used for big data processing. Python-based visualizations helped convert processed data into meaningful insights.

The project highlights how big data technologies can support data-driven decision-making in the automobile industry.